

b) The codomain of the transformation $x \mapsto Ax$ is the set of all linear combinations of the columns of A .

Sol: On page 65 under the section "Matrix Transformations", it is mentioned that the codomain of T is $\mathbb{R}^m$ when each column of A has m entries. The range of T is the set of all linear combinations of the columns of A . It's not necessary that the range and the codomain are the same.

∴ The statement is False.

c) If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation and if c is in $\mathbb{R}^m$, then a uniqueness question is "Is c in the range of T ?"

Sol: From the solution of Example 1(d) on page 65, the question of whether c is in the range of T is an existence problem. ∴ The given statement is False.

d) A linear transformation preserves the operations of vector addition and scalar multiplication.

Sol: The given statement is True. This is mentioned as the first sentence on page 67.

e) The superposition principle is a physical description of a linear transformation.

Sol: From page 67, just above Example 4, if one reads it, they will realize that the statement is True.

24) Suppose vectors $v_1, \dots, v_p$ span $\mathbb{R}^n$, and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation. Suppose $T(v_i) = 0$ for $i = 1, \dots, p$. Show that T is the zero transformation. That is, show that if x is any vector in $\mathbb{R}^n$, then $T(x) = 0$.

Sol: By defn of Span, for any arbitrary vector x in $\mathbb{R}^n$, there exists scalars $c_1, c_2, c_3, \dots, c_p \in \mathbb{R}$ such that

$$c_1 v_1 + c_2 v_2 + \dots + c_p v_p = x$$

$$T(x) = T(c_1 v_1 + c_2 v_2 + \dots + c_p v_p) = T(c_1 v_1) + T(c_2 v_2) + \dots + T(c_p v_p)$$

$$= c_1 T(v_1) + c_2 T(v_2) + \dots + c_p T(v_p) \quad (\text{Condition 2})$$

$$= c_1 \cdot 0 + c_2 \cdot 0 + \dots + c_p \cdot 0 \quad (\text{Given}) = 0 + 0 + 0 + \dots + 0 = 0 \quad (\text{Algebraic prop (ii)})$$

□